Chronic Kidney Disease

SOURCE-NCDC

Chronic kidney disease (CKD) is a condition in which the kidneys are damaged or cannot filter blood as well as healthy kidneys. Because of this, excess fluid and waste from the blood remain in the body and may cause other health problems.

Fast Stats • 30 million people or 15% of US adults are estimated to have CKD.* • 48% of those with severely reduced kidney function but not on dialysis are not aware of having CKD. • Most (96%) people with kidney damage or mildly reduced kidney function are not aware of having CKD.

Risk Factors for Developing CKD Adults with diabetes, high blood pressure, or both have a higher risk of developing CKD than those without these diseases. Other risk factors for CKD include heart disease, obesity, and a family history of CKD.

Symptoms, Testing, and Treatment • People with CKD may not feel ill or notice any symptoms. The only way to find out for sure if you have CKD is through specific blood and urine tests. These tests include measurement of both the creatinine level in the blood and protein in the urine. • Once detected, CKD may be addressed through lifestyle changes, including making healthier choices about what you eat and drink, and can often be treated with medications. These approaches and treatments may keep CKD from getting worse and may prevent additional health problems such as heart disease. • People with diabetes or high blood pressure who are diagnosed with CKD should talk to their doctor about treating these conditions to keep their blood sugar and blood pressure under control and lower their risk for kidney failure.

Health Problems Caused and Affected by CKD Kidney Failure • Kidney disease usually gets worse over time though treatment has been shown to slow progression. When the kidneys stop working, dialysis or kidney transplant is needed for survival. Kidney failure treated with dialysis or kidney transplant is called end-stage renal disease (ESRD). Not all patients with kidney disease progress to kidney failure and, in some patients, kidney disease progresses to kidney failure even with proper treatment.

Heart Disease and Stroke • Having kidney disease increases the chances of also having heart disease and stroke. • Managing blood pressure, blood sugar, and cholesterol levels—all risk factors for heart disease and stroke—is more difficult, but much more important in the presence of CKD. Other Health Consequences of CKD • Anemia or low number of red blood cells can cause fatigue and weakness. • Infections can occur because of a weakened immune system. • Low calcium levels and high phosphorus levels in the blood can cause bone problems. • High potassium levels in the blood (hyperkalemia) can cause an irregular or abnormal heartbeat. • Loss of appetite or eating less. • Excess fluids in the body causing high blood pressure, swelling in the legs, or shortness of breath because of fluid in the lungs (a condition known as pulmonary edema). • Depression or lower quality of life. Risk of Dying Premature death from both heart disease and from all causes is higher in adults with CKD compared with adults without CKD.

Opportunities to Prevent CKD and Lower the Risk for Kidney Failure • Control risk factors for CKD that can be modified. – High blood pressure. – High blood sugar levels. • Test for kidney disease among people who are at high risk for developing CKD. – Testing people with diabetes or with high blood pressure has been shown to be a cost-effective way of identifying people with CKD. • Manage CKD. – Make lifestyle changes (e.g., healthy eating) to prevent more kidney damage. – Use medications (e.g., drugs to lower blood pressure) to slow CKD progression. – Avoid conditions or exposures that

can harm the kidneys or cause a sudden drop in kidney function (called acute kidney injury) and may quicken CKD progression. • Kidney infections. • Medications. Over-the-counter pain medicines like ibuprofen and naproxen.

 Certain antibiotics. • Herbal supplements. • Dyes that are used to make the blood vessels or organs visible on X-rays or other imaging tests

SOURCE-WHO

- Kidney disease includes acute kidney injury (AKI) and chronic kidney disease (CKD).
- Simple blood (serum creatinine) and urine tests (for albuminuria) used in primary health care can detect CKD.
- An estimated 674 million people have chronic kidney disease worldwide; most reside in low- and middle-income countries.
- The most severe form of CKD is kidney failure, which requires dialysis or kidney transplantation to sustain life.
- The global burden of acute kidney injury (AKI) is unknown.

Overview

Kidney disease occurs when the kidneys can no longer remove waste and excess fluid from the bloodstream normally. Acute kidney injury (AKI) has an abrupt onset and is often reversible with timely intervention. In contrast, chronic kidney disease (CKD) progresses gradually and is frequently irreversible, with severity ranging from mild dysfunction to kidney failure. People with kidney failure typically require dialysis or kidney transplantation to survive.

Kidney disease causes substantial morbidity, disability, and premature mortality, in part because it causes and is caused by cardiovascular disease. The burden of kidney disease is rising in parallel with diabetes, hypertension and population ageing. Global inequities in access to testing, essential medications, health- care workers and kidney replacement therapies are a key global challenge and contribute to millions of preventable deaths each year.

Symptoms

Kidney disease is often asymptomatic until late stages, which is why regular testing is important for people at risk. In advanced or severe kidney disease, symptoms may include fatigue, breathlessness, generalized itchiness, leg swelling, muscle cramps, nausea or vomiting.

Chronic kidney disease (CKD)

CKD has a wide range of causes, including diabetes, hypertension, cardiovascular disease, glomerulonephritis (inflammation of the kidney's filtering units), some genetic conditions, certain drugs and toxins and infections. In low- and middle-income countries (LMICs), a substantial proportion of people with CKD do not have a known exposure to known risk factors.

CKD is diagnosed by using serum creatinine results to estimate glomerular filtration rate (eGFR); low eGFR suggests the presence of kidney disease. Two values of eGFR <60 ml/min/1.73m^2 obtained at least 90 days apart indicate that CKD is present. Persistently elevated urinary albumin excretion

(“albuminuria”) is identified by urinary albumin-creatinine ratio (ACR) of >3 mg/mmol (30 mg/g) and also indicates the presence of CKD.

The risk of developing CKD can be reduced by a healthy lifestyle, including regular physical activity, maintaining a healthy body weight and refraining from tobacco use. In people with diabetes, hypertension and cardiovascular disease, good control of blood pressure, blood glucose and blood lipids, alongside these lifestyle measures, can further help to reduce the risk of developing CKD.

Integrating kidney health into primary care is essential for early diagnosis and timely treatment. Routine testing (“case-finding”) with eGFR and albuminuria can be arranged in primary care for people with hypertension, diabetes and cardiovascular disease, aiming to diagnose and treat kidney disease at an early stage.

The management of CKD focuses on slowing progression, reducing cardiovascular risk, and preventing complications of advanced kidney disease.

For treatment of kidney failure, kidney replacement therapy (dialysis or kidney transplantation) may be required.

Where transplantation or dialysis is unavailable, unaffordable, or not desired by the individual, conservative kidney management provides supportive care focused on symptom control and quality of life.

Acute kidney injury (AKI)

AKI represents a deterioration of kidney function over hours to days and can be reversible with timely care. Although AKI and CKD differ in onset and reversibility, they are closely connected: AKI can lead to CKD, and CKD increases the risk of AKI. Even a single episode of AKI raises the risk of CKD and death.

Common causes of AKI include sepsis, major surgery, trauma, complications of pregnancy, nephrotoxic medications and volume depletion.

AKI is diagnosed by rising serum creatinine levels above baseline (pre-illness) values, and/or abnormally low levels of urine output.

AKI can be prevented by avoiding and treating dehydration, infections and pregnancy-related complications.

Management of AKI focuses on treating the cause and correcting fluid/electrolyte disturbances and providing temporary dialysis when needed.